PROJECT REPORT

Secure and Scalable AWS VPC Network with VPC Peering

AWS cloud networking, private workloads, database access, and inter-VPC connectivity

Prepared by	Huzaif Baig
Project area	AWS Cloud Networking
Report date	November 9, 2026

Submitted as a practical cloud networking project report

1. Project Details

Project title	Secure and Scalable AWS VPC Network with VPC Peering
Author	Huzaif Baig
Platform	Amazon Web Services (AWS)
Core services	Amazon VPC, EC2, subnets, route tables, Internet Gateway, NAT Gateway, and VPC Peering

Project summary

This project documents the design and configuration of an AWS virtual network with public and private subnets. It covers controlled internet routing, EC2-based access to a private database workload, and VPC peering to connect separate VPC networks. The repository includes AWS console captures, terminal tests, and architecture diagrams.

2. Objective

The objective is to build and verify a segmented AWS network that separates internet-facing resources from private workloads, provides controlled network paths for administration and outbound access, and enables private communication between VPCs through VPC Peering.

• Create a custom VPC and organize resources into public and private subnets.

• Configure route tables and gateways so traffic follows the intended network path.

• Run a public EC2 instance as a controlled administrative entry point and keep the private instance off the public internet.

• Demonstrate connectivity to a database hosted in the private network.

• Establish VPC peering and validate communication between the connected networks.

3. Methodology

1. Plan the network. Choose a VPC address range and divide the network into public and private subnet roles.

2. Build VPC resources. Create the VPC, subnets, route tables, and associations; attach an Internet Gateway for public-subnet routing.

3. Provide controlled outbound access. Configure a NAT Gateway in the public subnet and route private-subnet outbound traffic through it, without assigning the private instance a public IP.

4. Deploy and access instances. Launch the public EC2 host and private workload. Use the public host as a bastion for administrative SSH access to the private instance.

5. Configure the database workload. Place the database in the private network and test access using its private address from an authorized network path.

6. Peer VPC networks. Create a peering request between the VPCs, accept it, and add routes for the peer network ranges to the appropriate route tables.

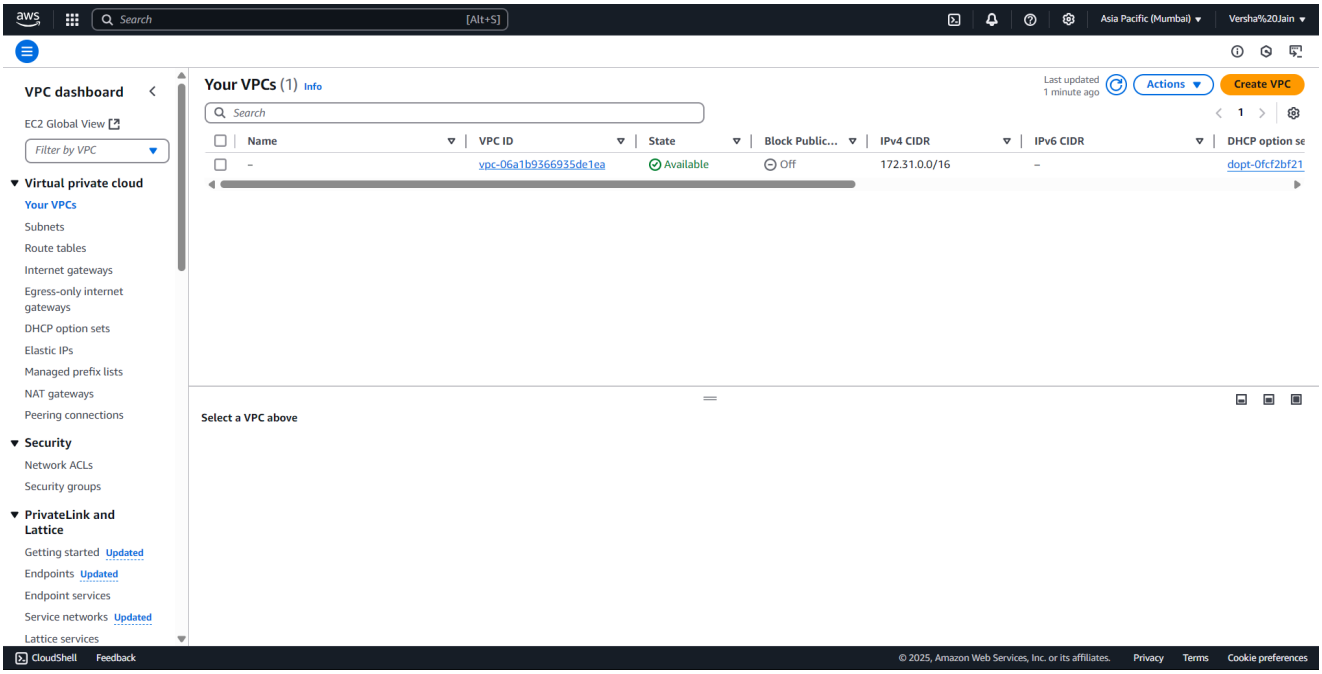
7. Verify and document. Run connectivity checks, record console and terminal evidence, and summarize the resulting network behavior.

Security note: NAT provides a route for outbound connections; it does not by itself make a private instance publicly reachable. Inbound access remains governed by routing, security groups, network ACLs, and the configured administrative path.

4. Screenshots and Descriptions

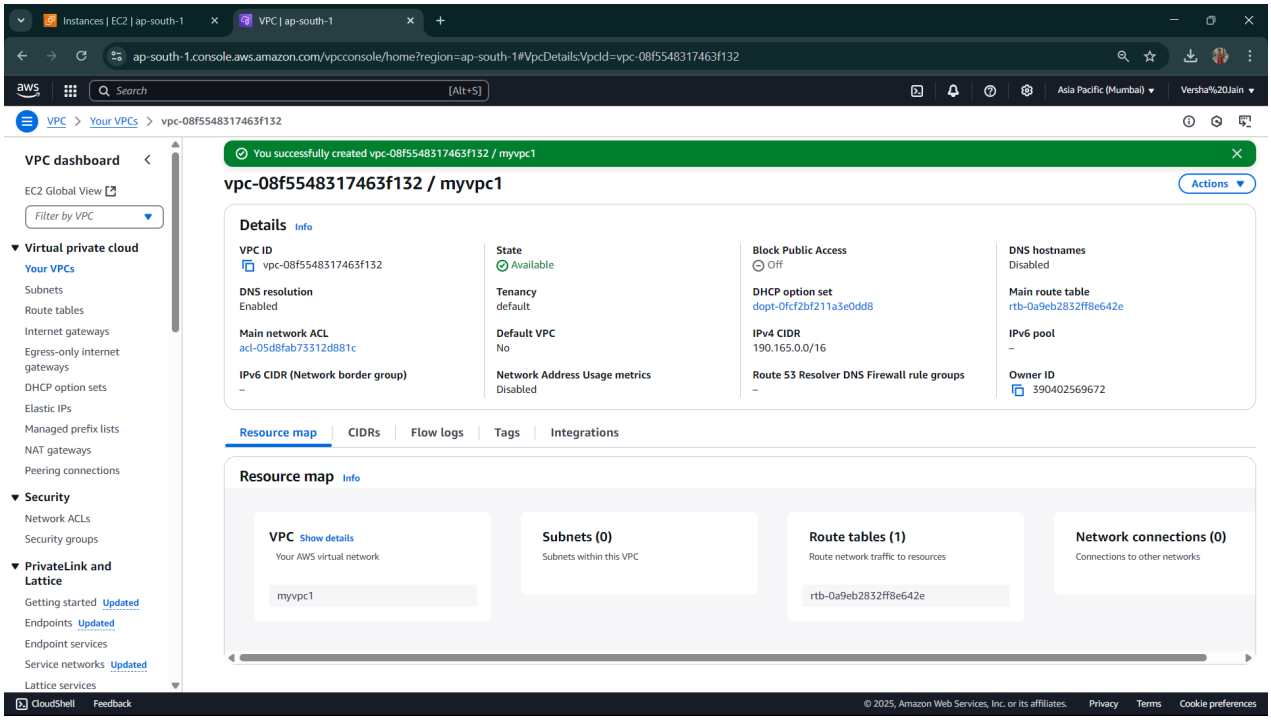
The following evidence section includes all 30 images supplied with the project. Captions identify each image by its role in the setup or verification workflow.

Figure 1. VPC console overview



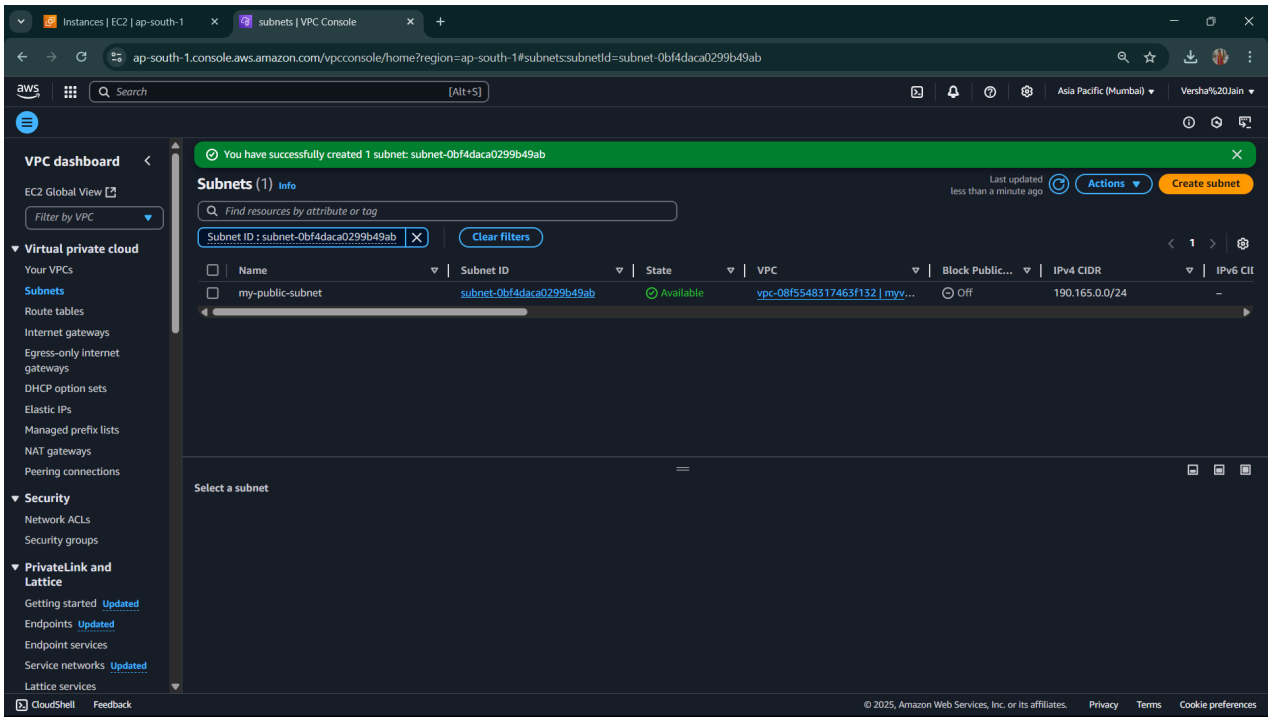
AWS VPC console view documenting the initial VPC setup and available network resources.

Figure 2. VPC configuration details



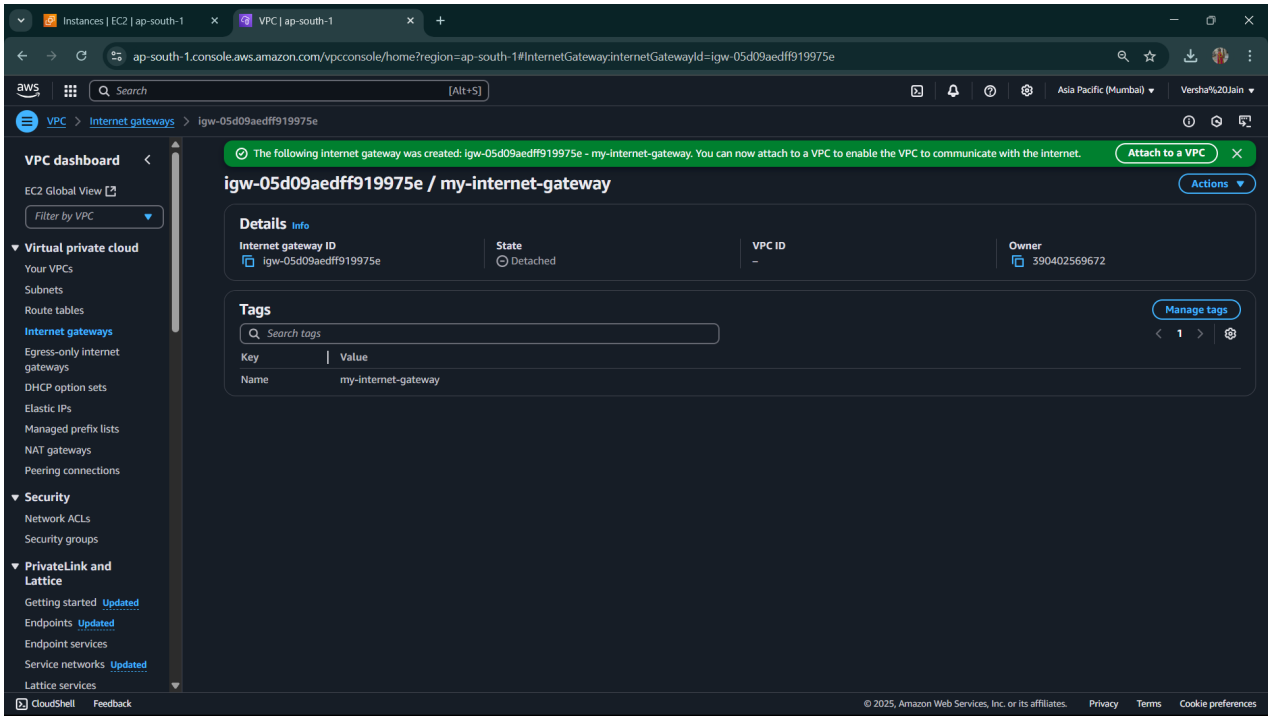
VPC details in the AWS console, including its identity and configured network settings.

Figure 3. Subnet configuration



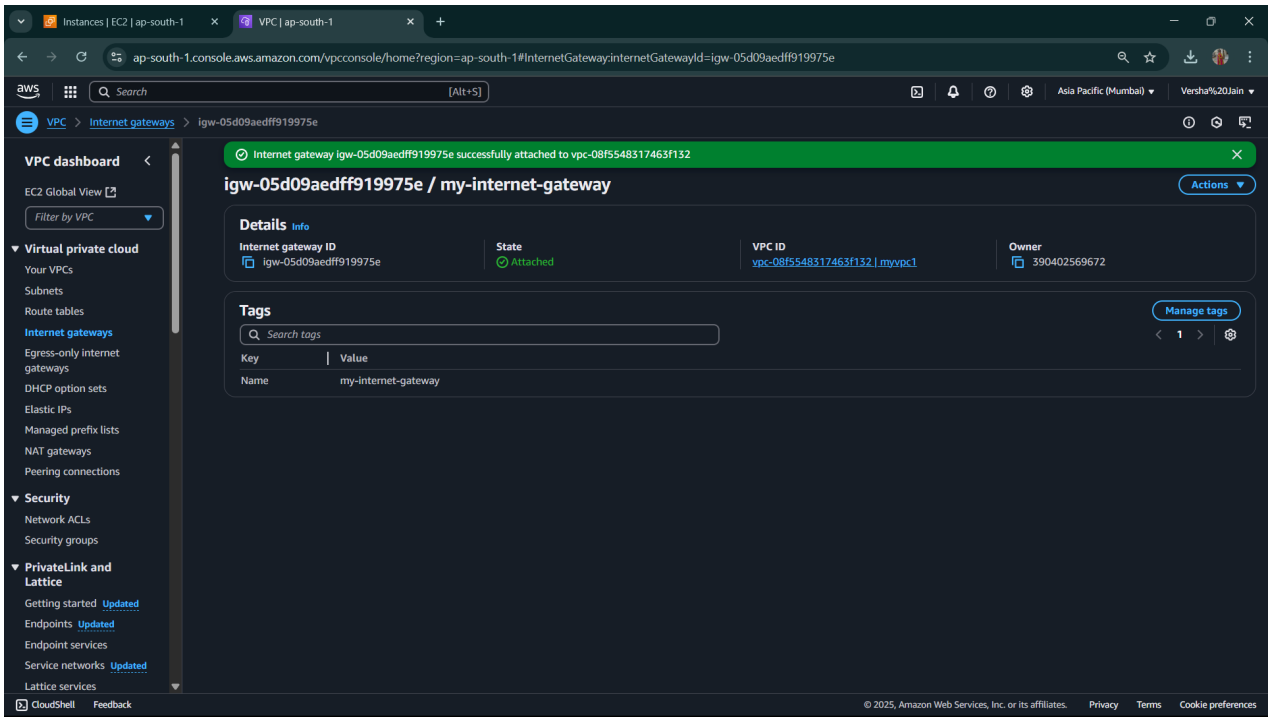
AWS console view used while configuring subnet resources for the VPC.

Figure 4. Subnet settings



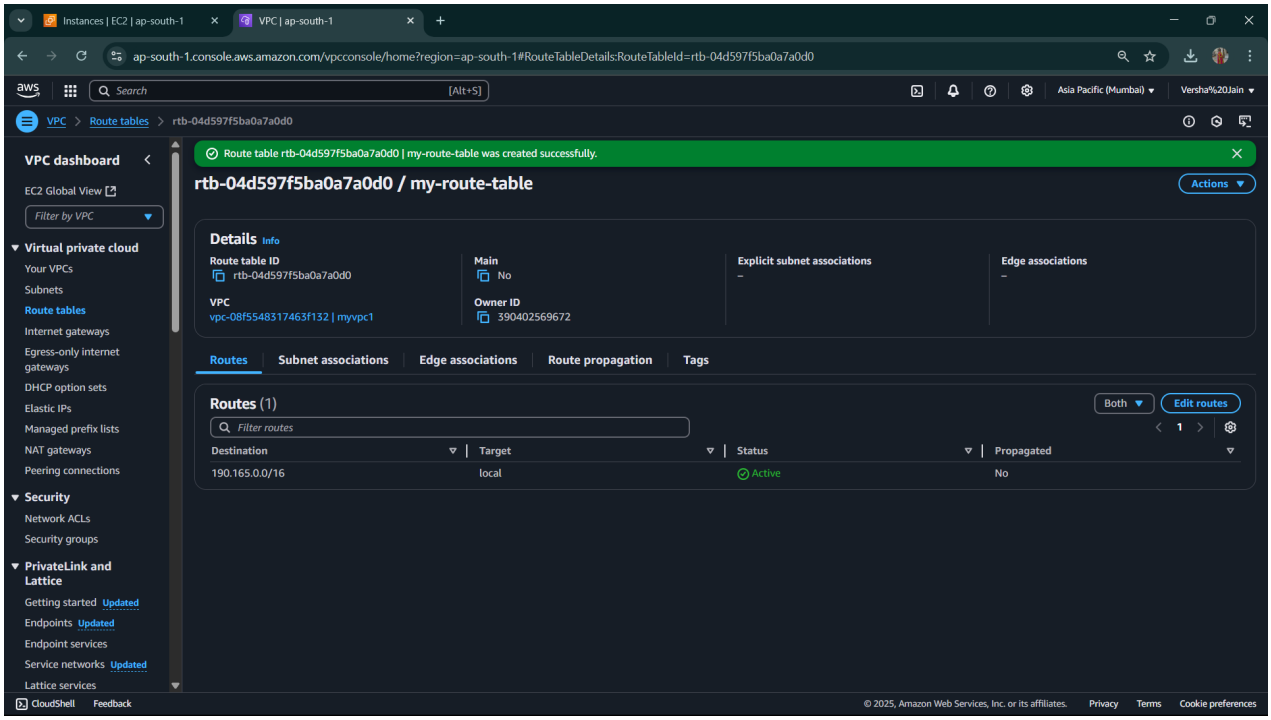
A further subnet configuration view showing the selected VPC networking context.

Figure 5. Subnet and network settings



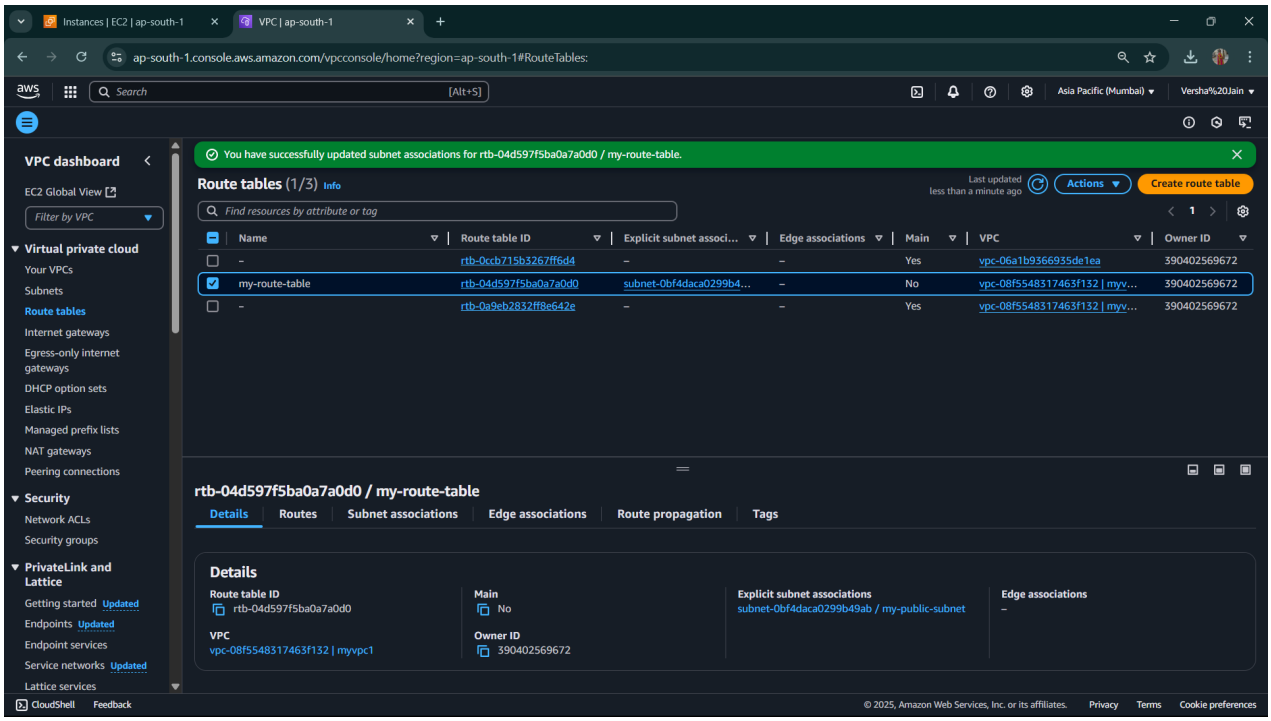
AWS console evidence from the subnet setup and associated networking workflow.

Figure 6. Route table configuration



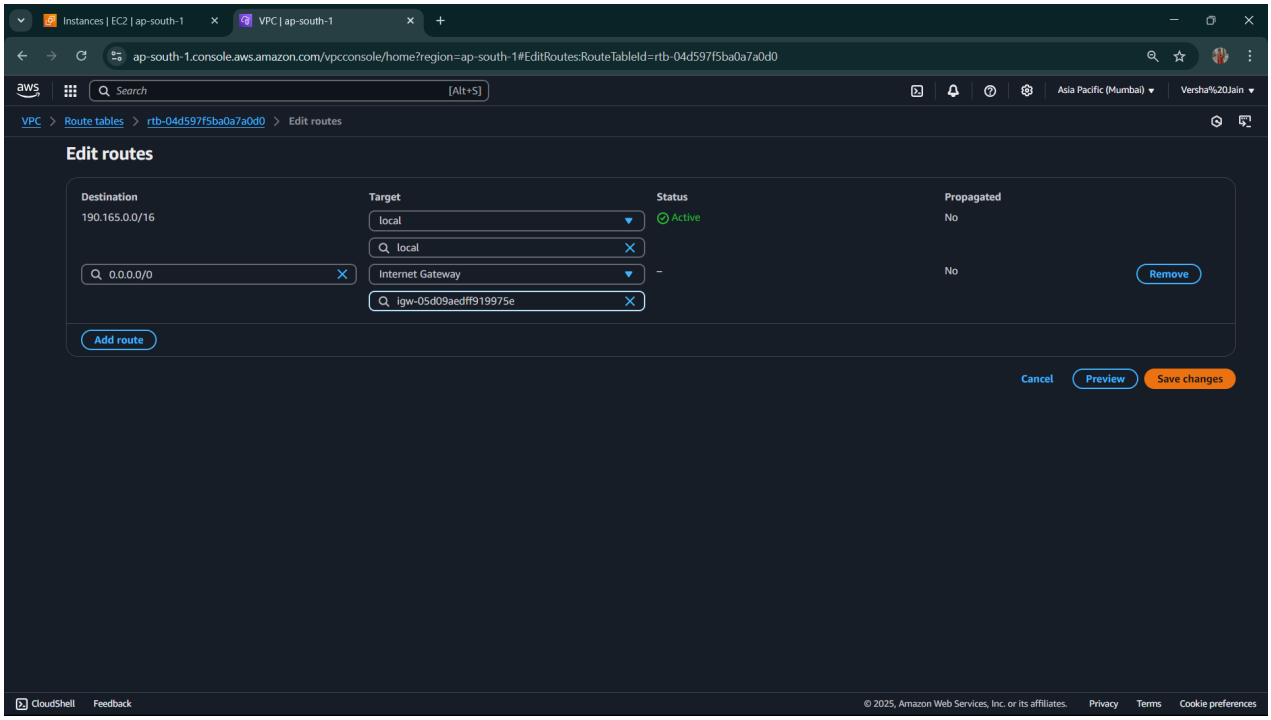
Route table details in the AWS console, supporting traffic control for the VPC subnets.

Figure 7. Route table associations



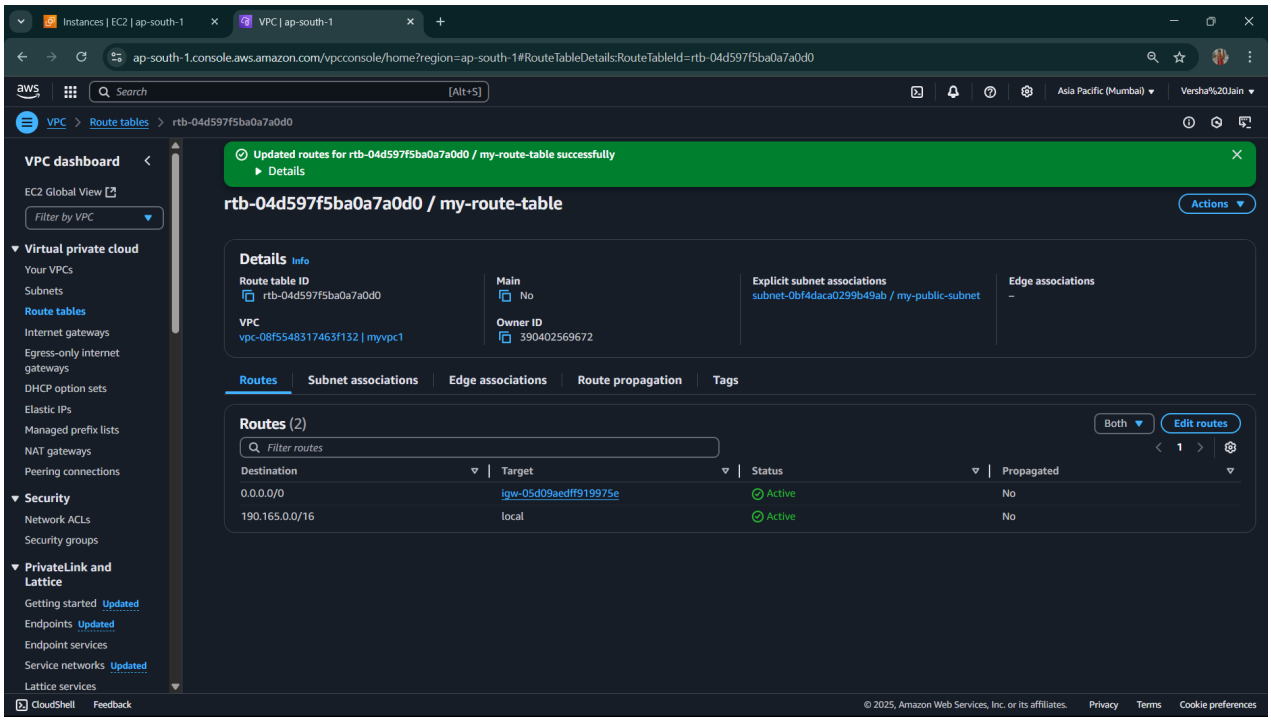
AWS console view of route table and subnet association settings.

Figure 8. Network resource setup



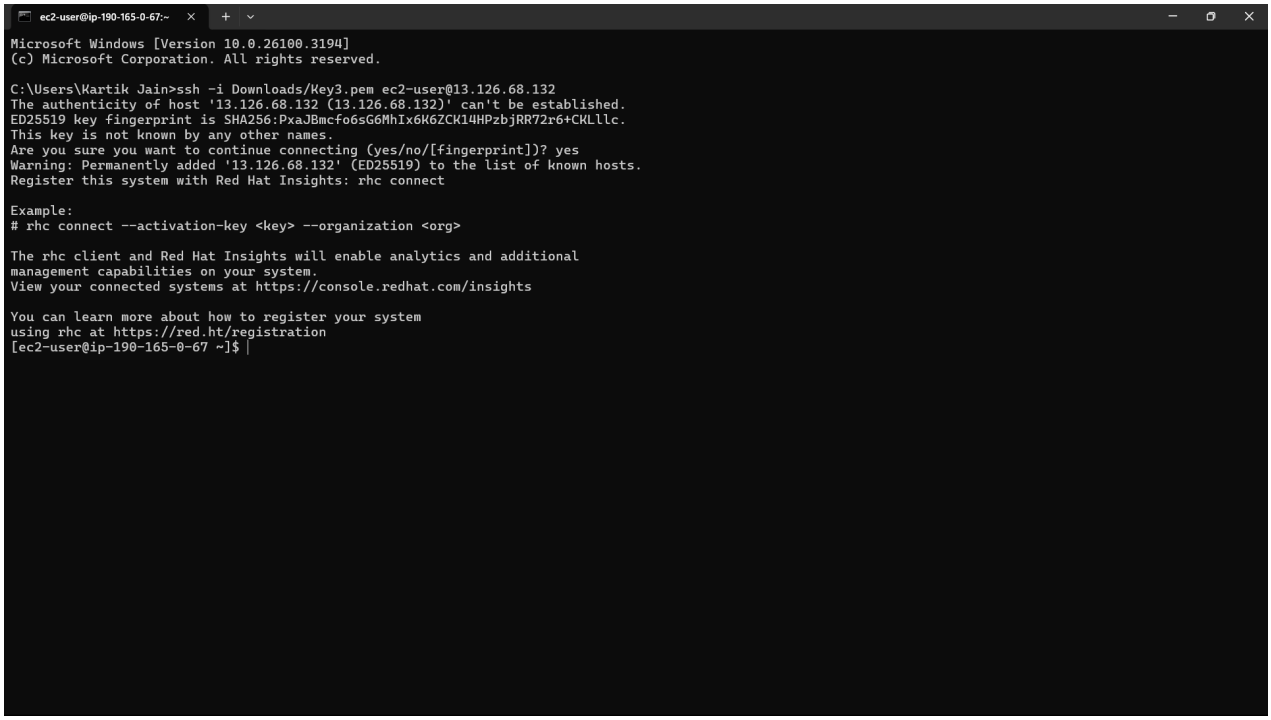
Configuration form used during setup of the VPC's supporting network resources.

Figure 9. VPC networking review



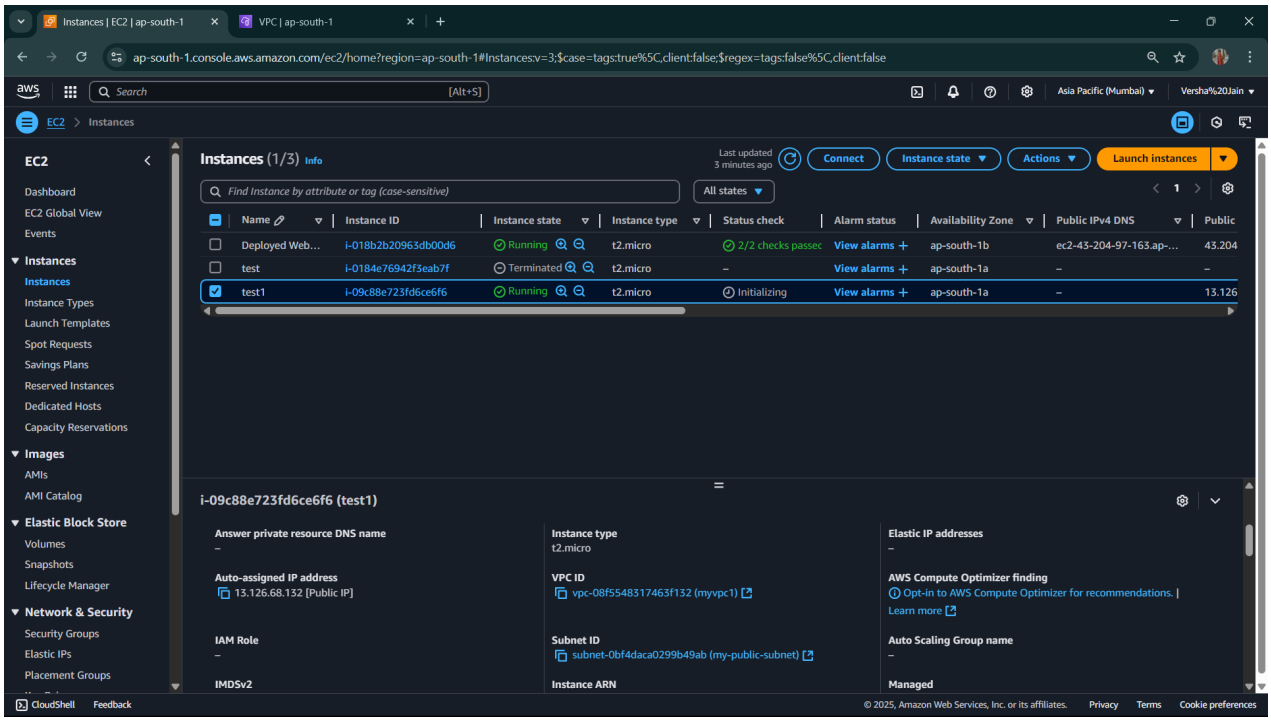
AWS console review of configured VPC networking resources and their state.

Figure 10. Command-line connectivity check



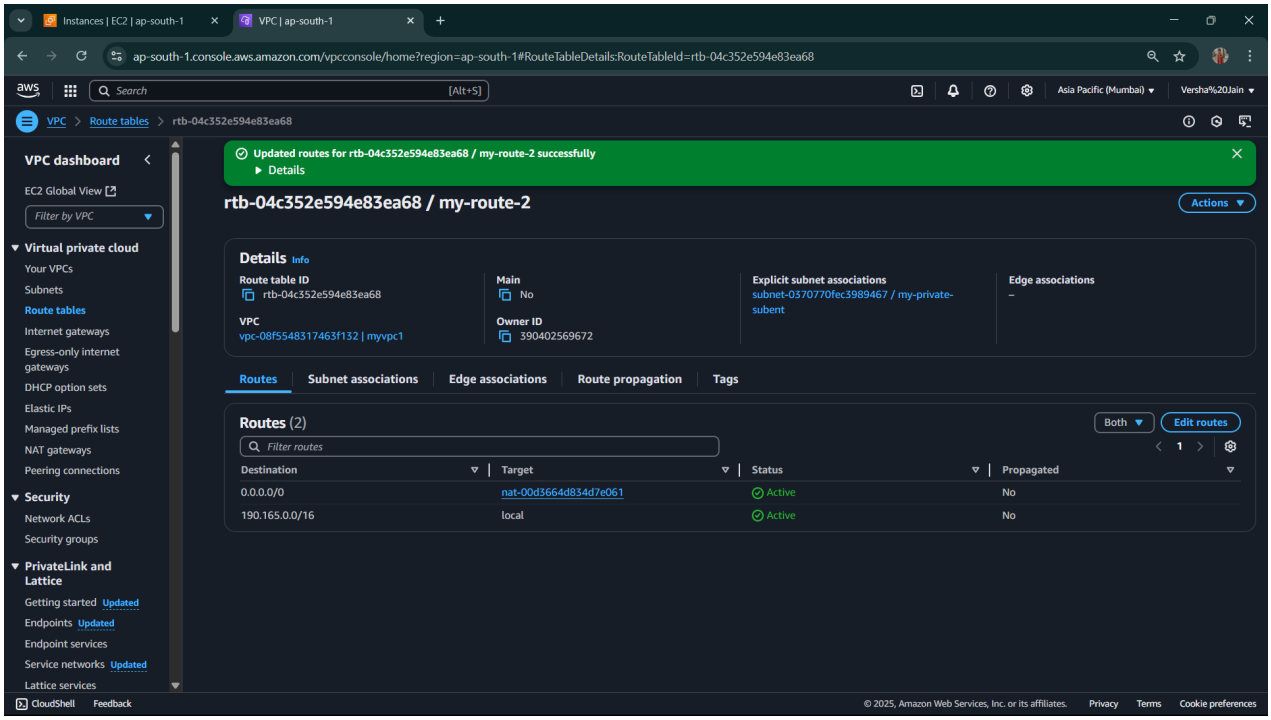
Terminal evidence from testing access and connectivity for the configured network.

Figure 11. Instance network interfaces



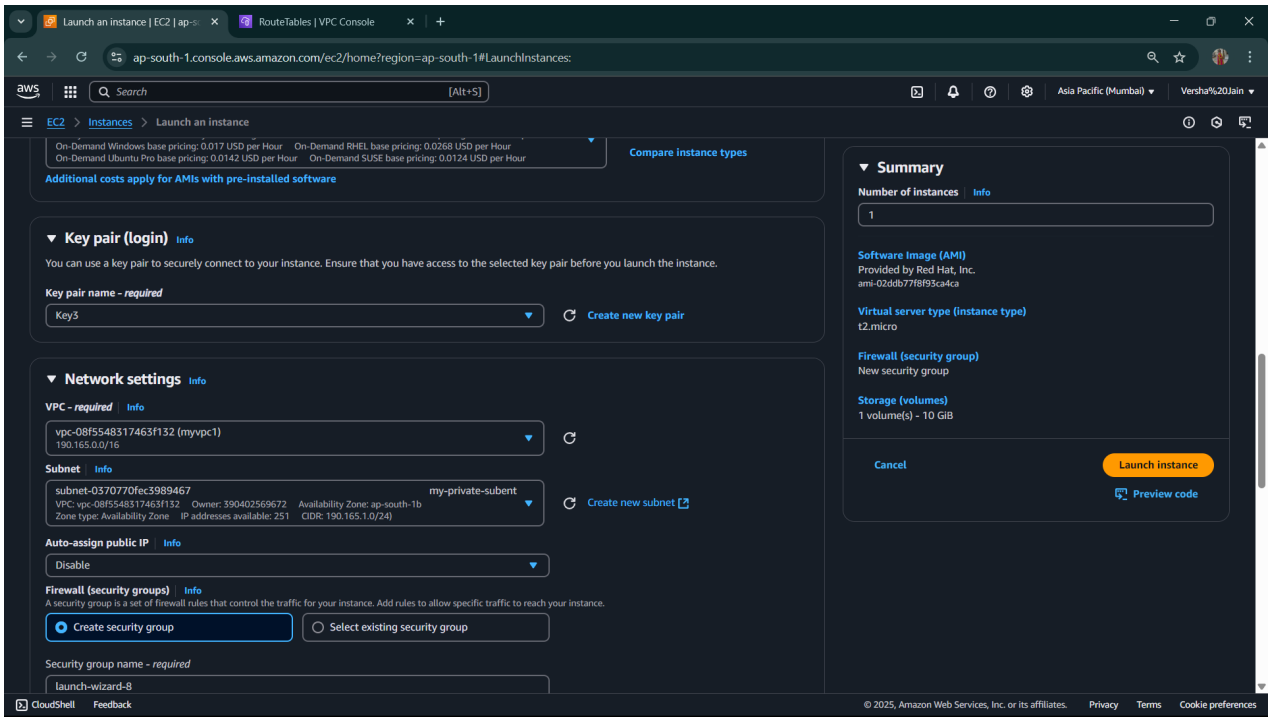
AWS console view of network interfaces and related instance connectivity resources.

Figure 12. VPC resource details



AWS console evidence reviewing the VPC and its related network configuration.

Figure 13. Gateway configuration



AWS console setup view for gateway and subnet routing resources.

Figure 14. Private network terminal test

```
C:\Users\Kartik Jain>ssh -i Downloads/Key3.pem ec2-user@13.126.68.132
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Thu Mar  6 09:03:01 2025 from 103.159.68.38
[ec2-user@ip-190-165-0-67 ~]$ sudo -i
[root@ip-190-165-0-67 ~]# vi /Key3.pem
[root@ip-190-165-0-67 ~]# lsNew] 27L, 1679B written
[root@ip-190-165-0-67 ~]# chmod 400 /Key3.pem
[root@ip-190-165-0-67 ~]# ssh -i /Key3.pem ec2-user@190.165.1.93
The authenticity of host '190.165.1.93 (190.165.1.93)' can't be established.
ED25519 key fingerprint is SHA256:mHUDR3lK5SW3DkPvX26dEHMVX7BkmbRo1qqYmQLo1Us.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '190.165.1.93' (ED25519) to the list of known hosts.
Register this system with Red Hat Insights: rhc connect

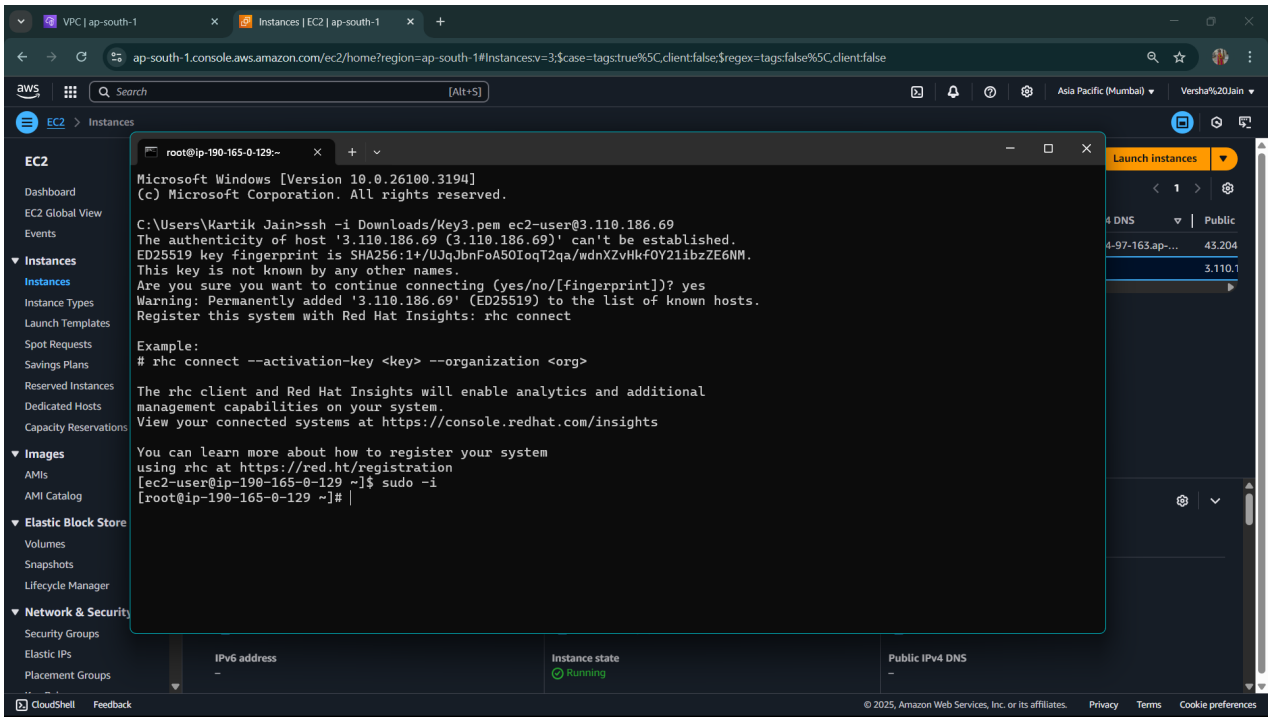
Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
[ec2-user@ip-190-165-1-93 ~]$
```

Terminal output captured during validation of the private network setup.

Figure 15. Instance access verification



Terminal session used to verify access to an instance as part of the VPC workflow.

Figure 16. VPC network layout

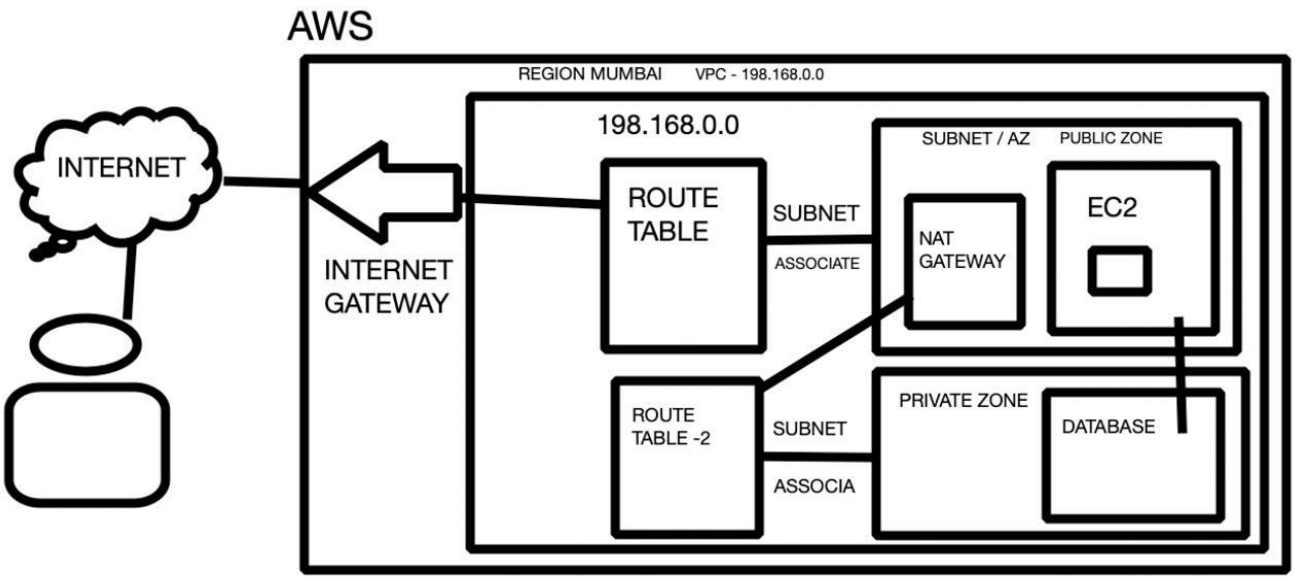


Diagram showing a VPC with public and private subnets, gateways, route tables, and workloads.

Figure 17. VPC and subnet overview

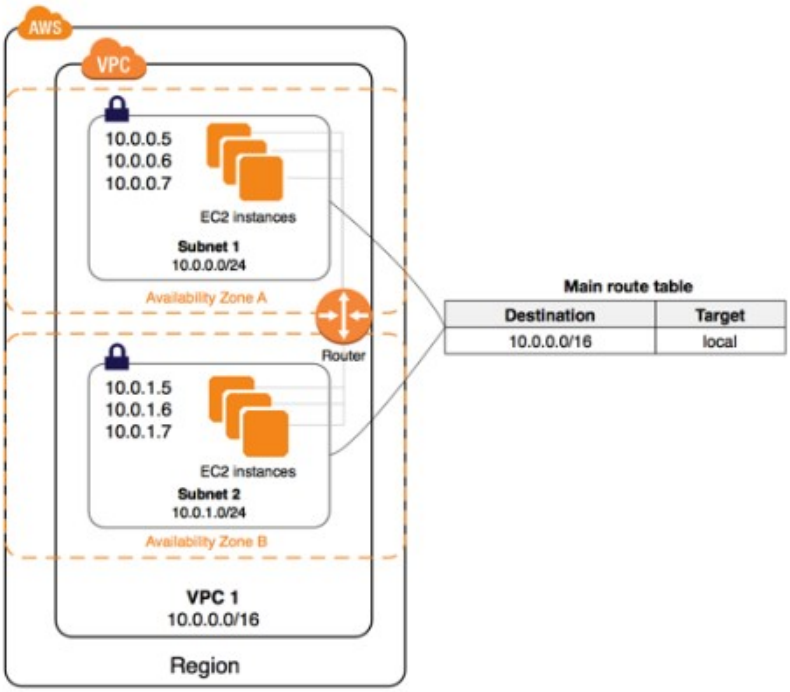
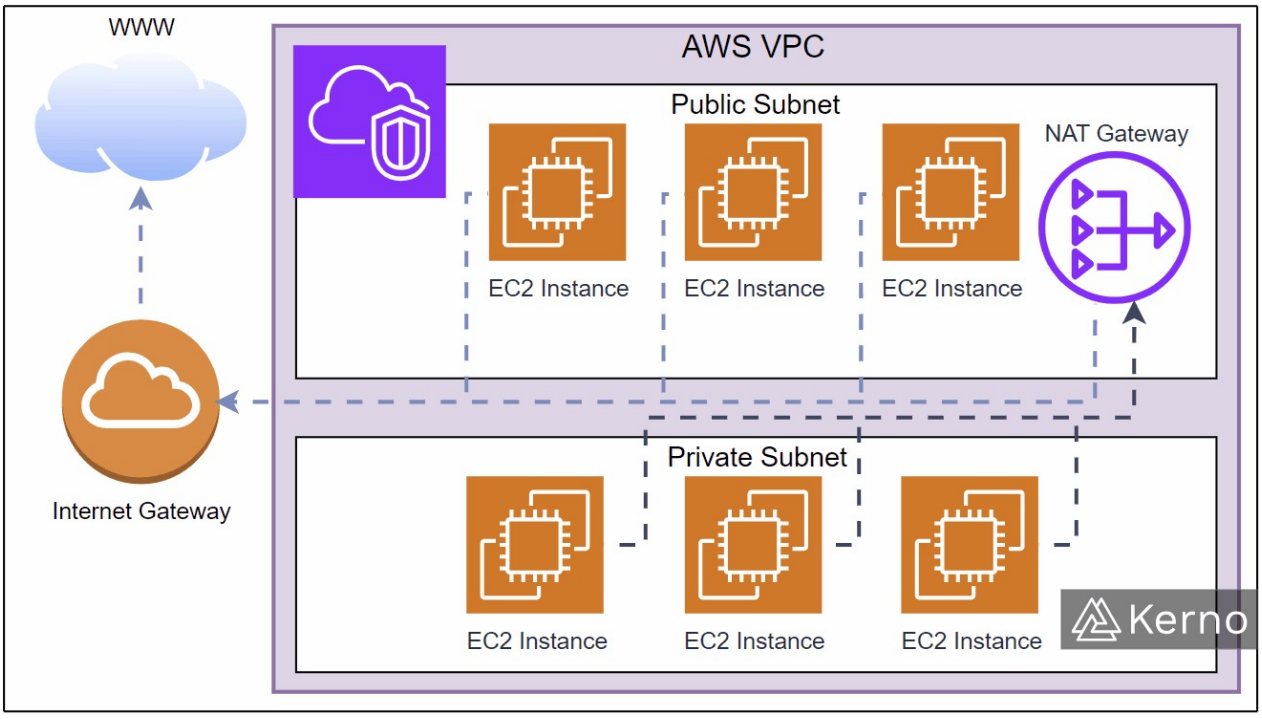


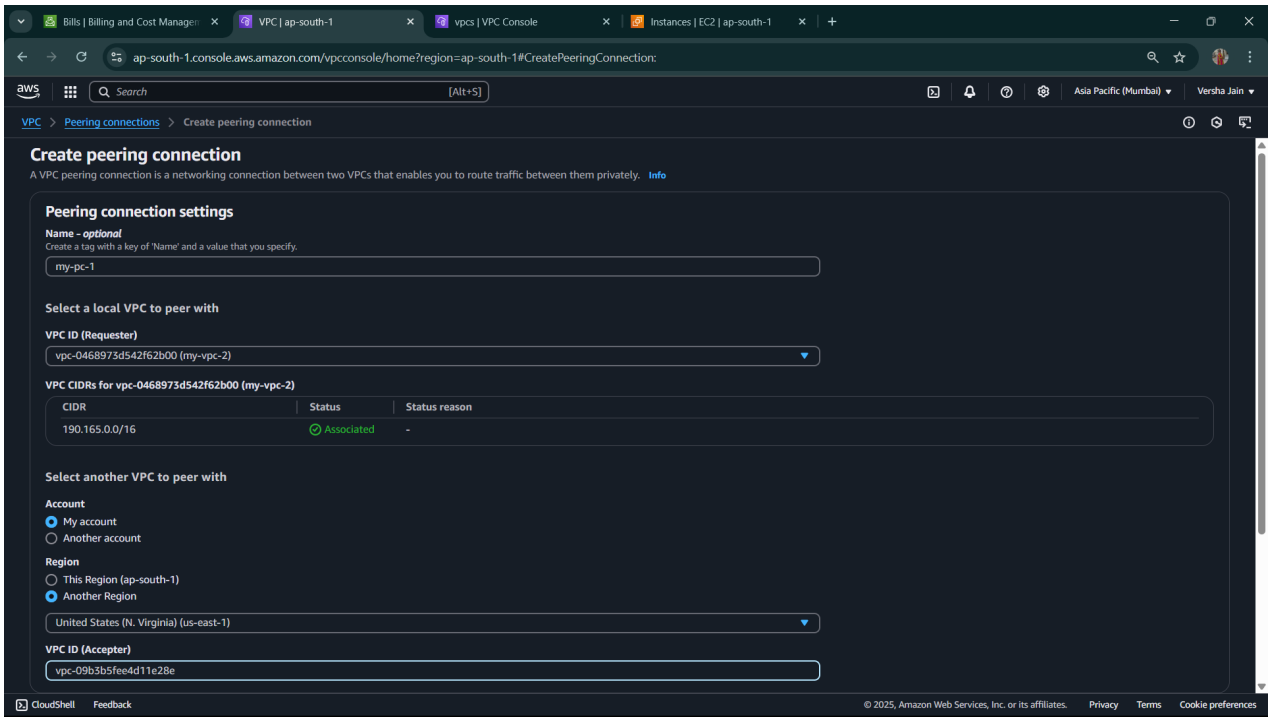
Illustration of the VPC address space, subnet separation, and gateway-based routing.

Figure 18. Public and private subnet architecture



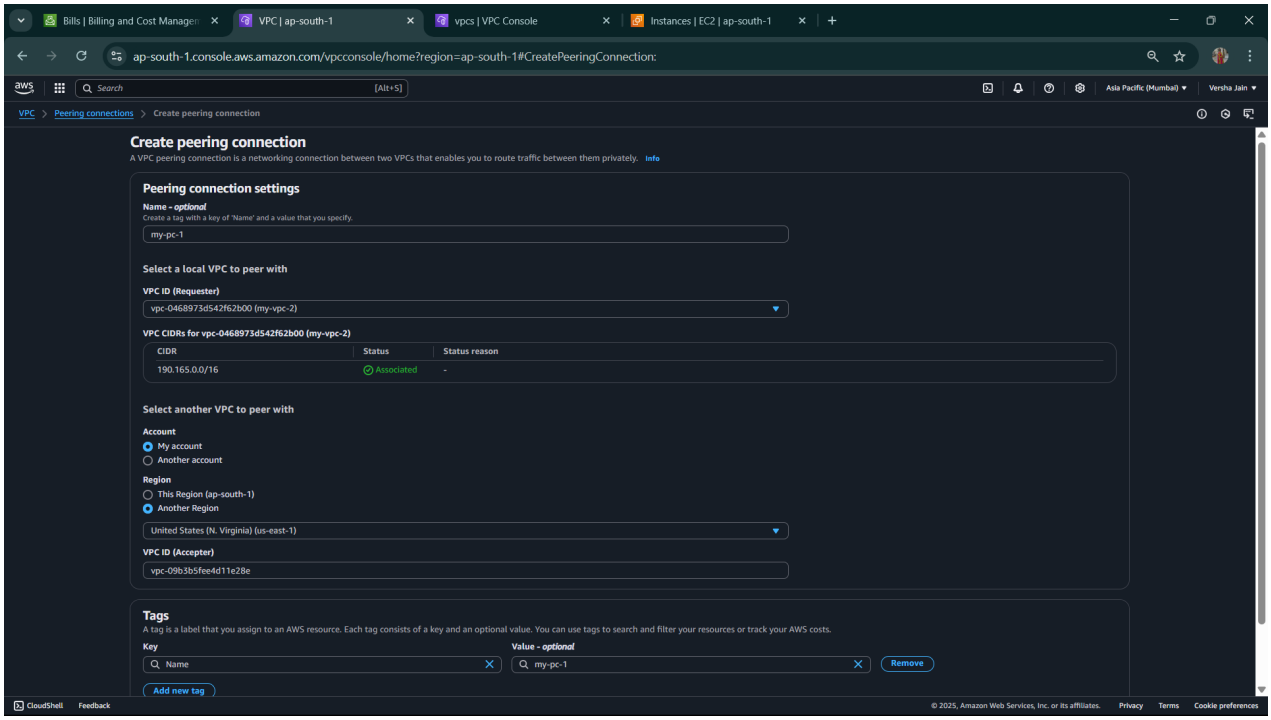
Architecture diagram showing public and private EC2 resources and a NAT gateway.

Figure 19. Peering connection setup



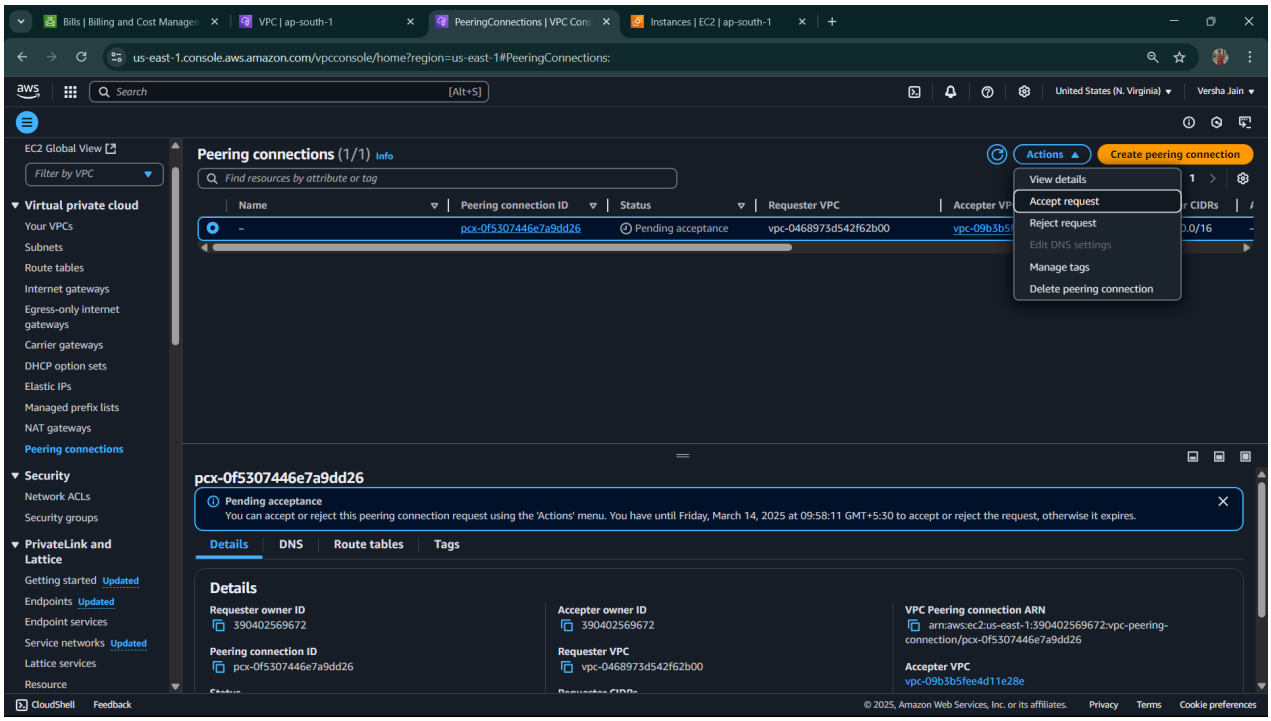
AWS console form for creating a VPC peering connection between requester and accepter VPCs.

Figure 20. Peering request configuration



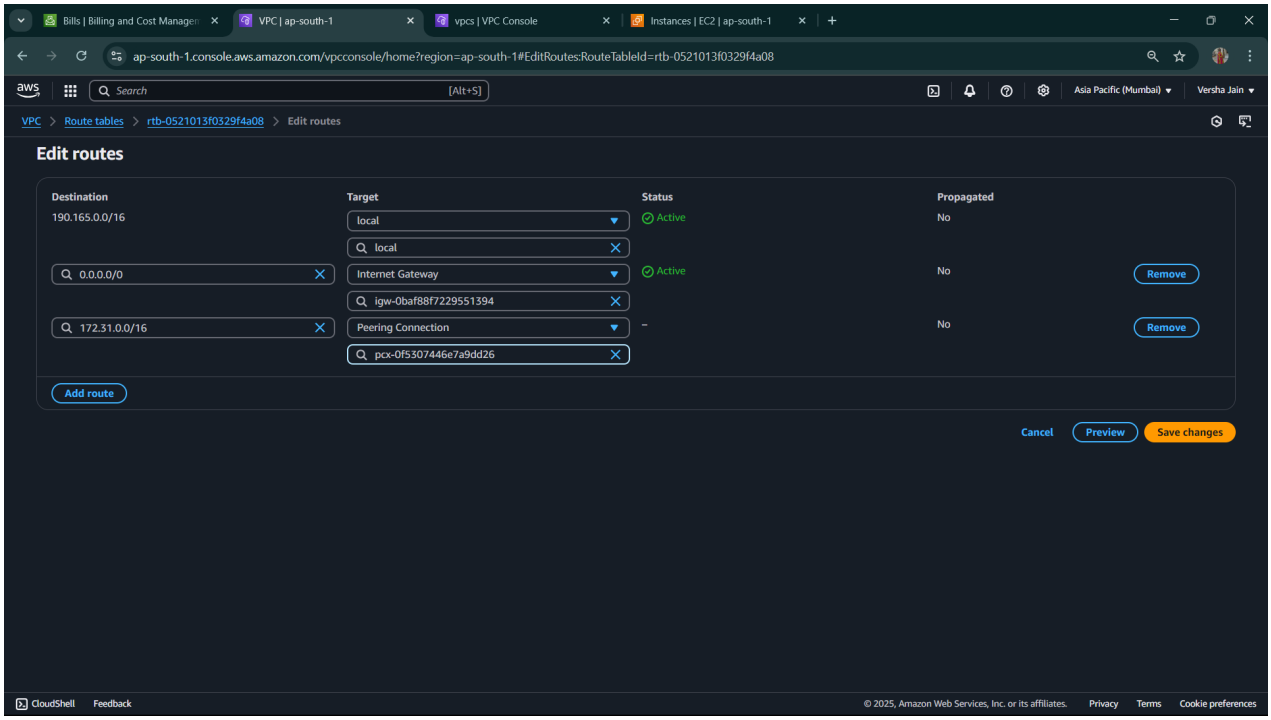
AWS console view of the VPC peering request and its selected network settings.

Figure 21. Peering connection details



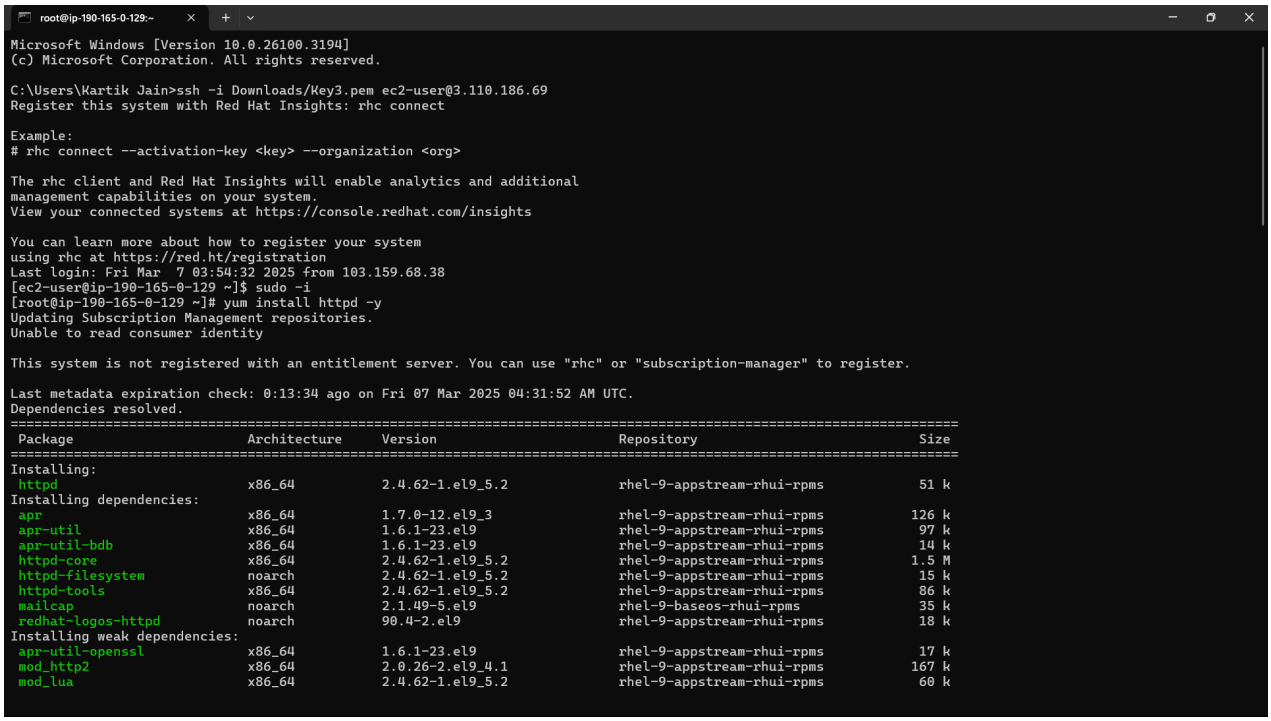
AWS console review of the peering connection and its status or available actions.

Figure 22. Peering route configuration



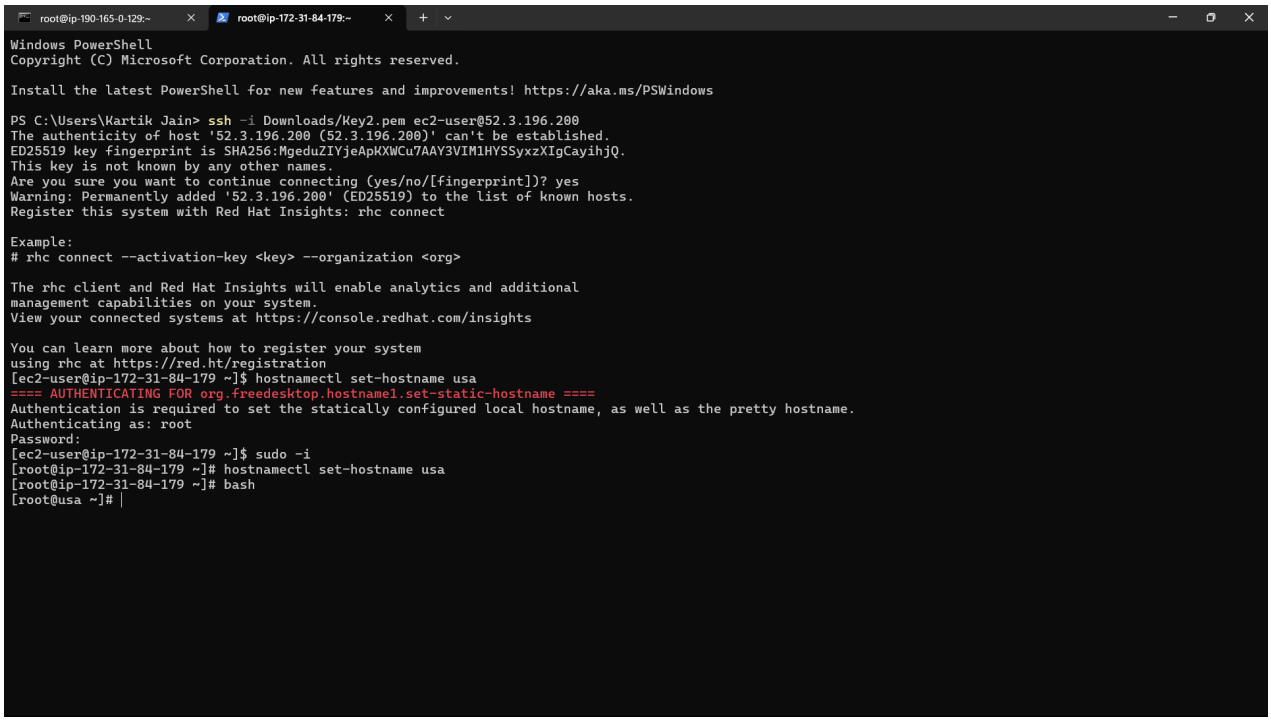
Route table configuration view used to direct traffic over the VPC peering connection.

Figure 23. Cross-VPC connectivity test



Terminal output documenting a connectivity test between network resources.

Figure 24. Peered instance access test



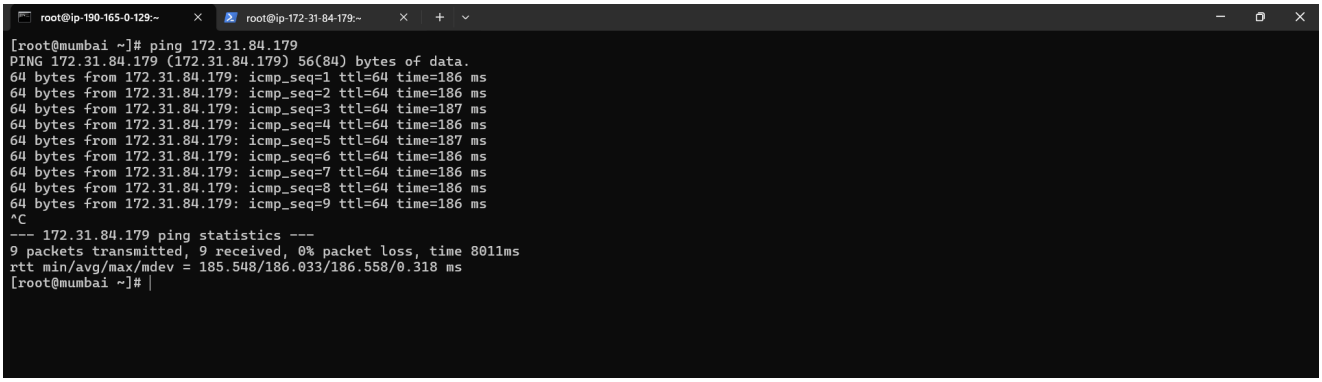
Terminal session showing commands used to test access across the peered networks.

Figure 25. Network reachability results



Terminal output captured while checking reachability between network endpoints.

Figure 26. Peering connectivity results



Terminal output documenting the VPC peering connectivity verification.

Figure 27. VPC peering concept

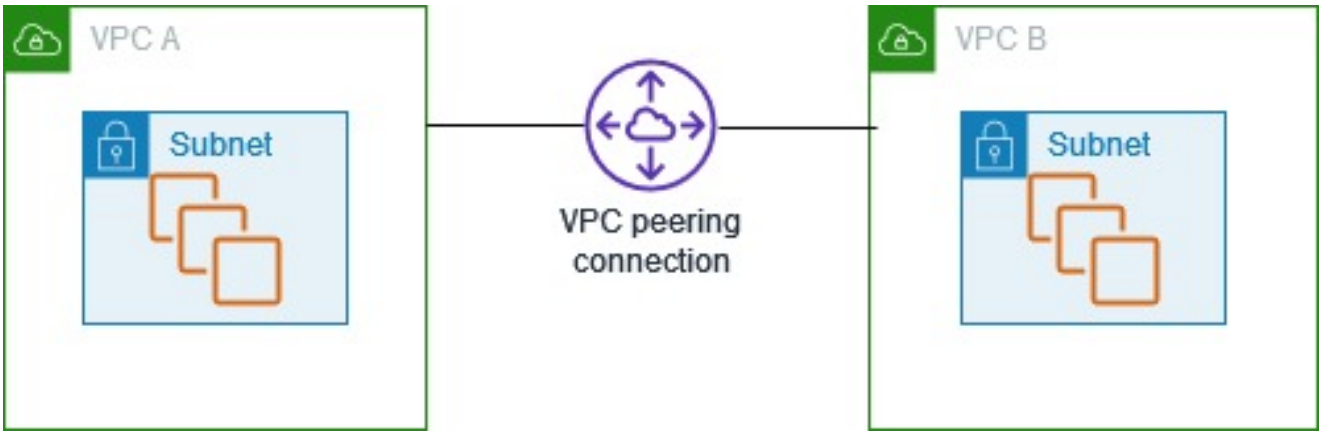


Diagram illustrating two VPCs connected by a VPC peering connection.

Figure 28. VPC peering network layout

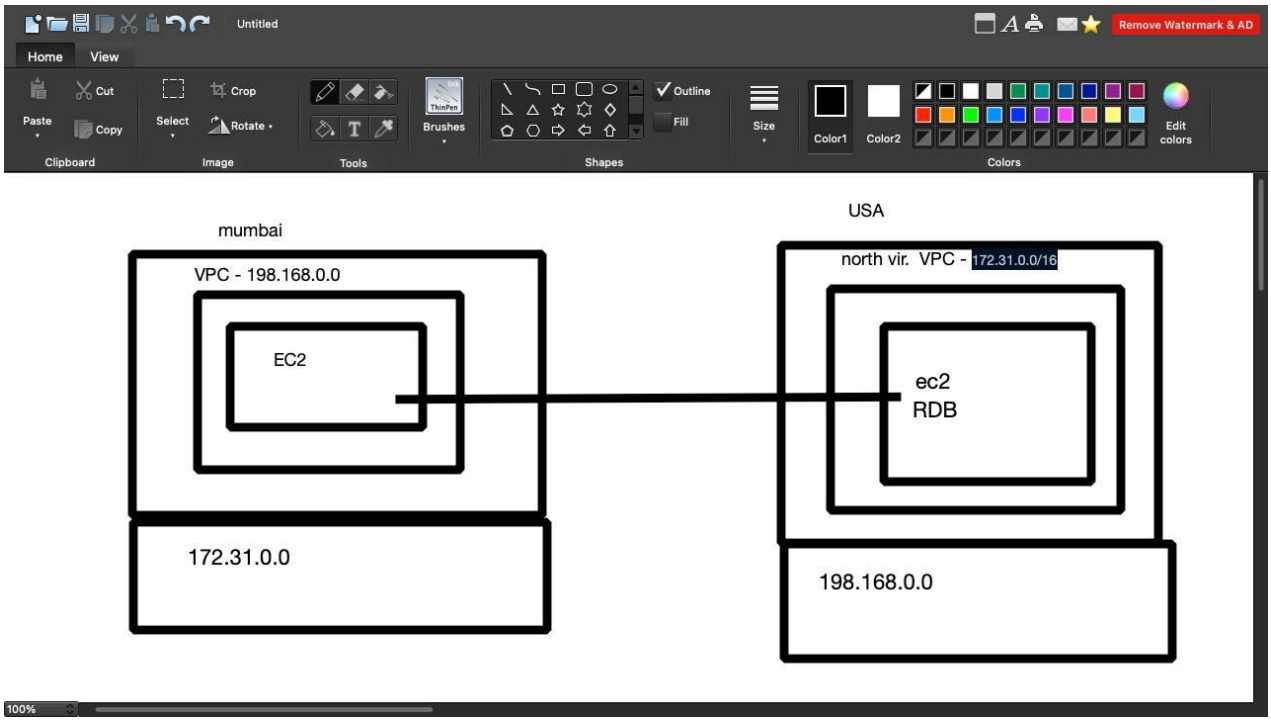
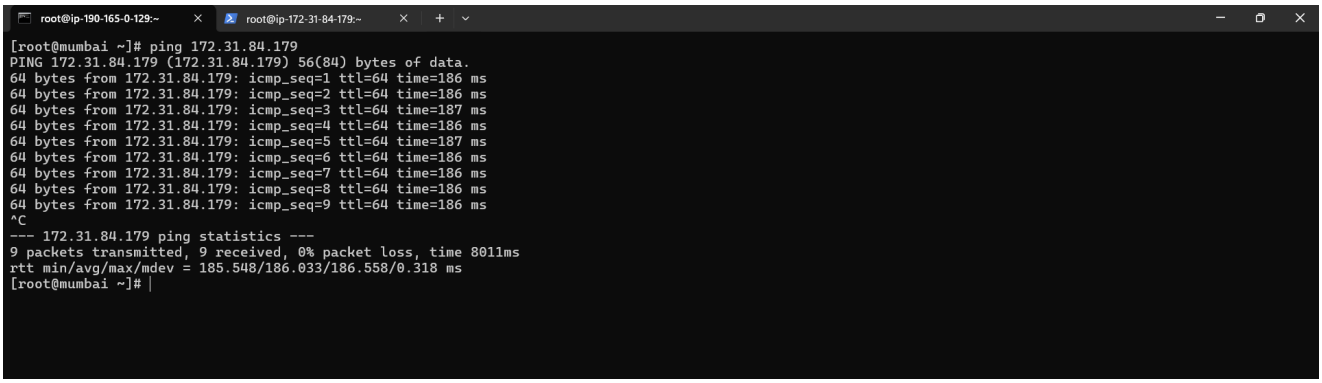


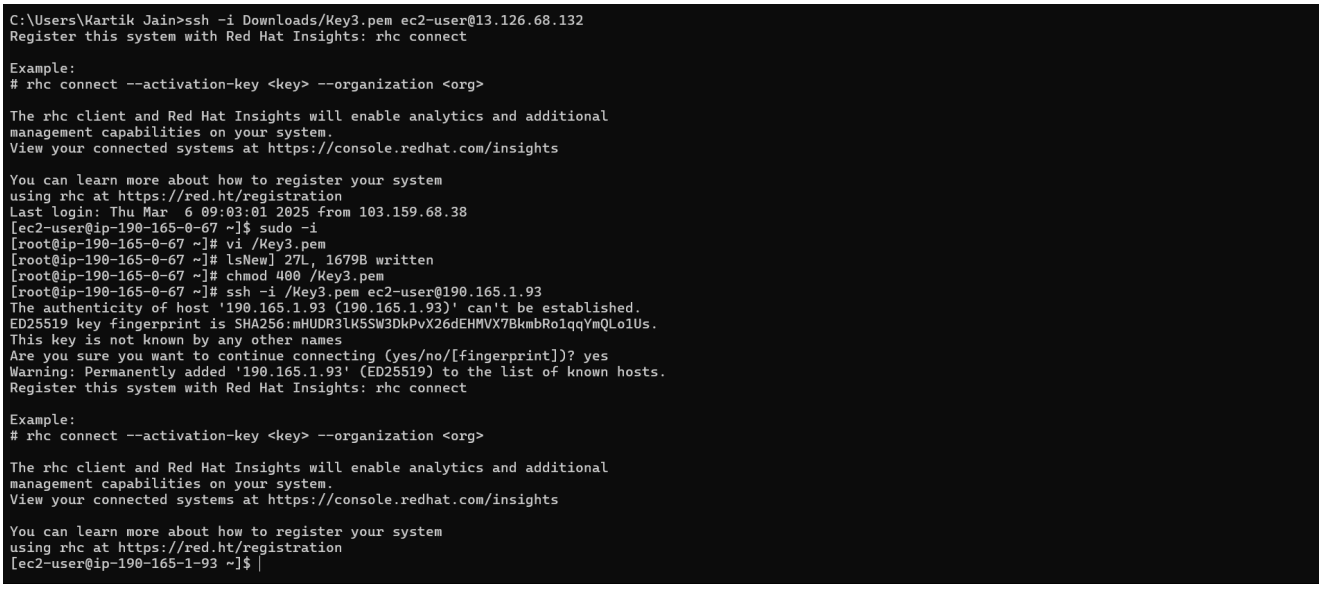
Diagram of two VPC networks and their private subnets connected through peering.

Figure 29. Private workload outbound connectivity



Terminal output associated with the documented internet-connectivity test for the private database workload.

Figure 30. Successful private database connection



Terminal evidence of a successful connection to the database hosted on a private instance.

5. Results and Conclusion

Results

The project materials document a VPC divided into public and private network areas, gateway and route-table configuration, and EC2-based access to a private workload. The terminal captures and README describe successful private database access and connectivity checks. The peering materials document a peering connection between VPCs and tests intended to verify communication across the connection.

Together, these configuration and test records demonstrate the central project outcome: a segmented AWS network with a private database workload and a network path for communication between VPCs. Actual reachability depends on matching routes, security-group rules, network ACLs, and the address ranges configured in the AWS account.

Conclusion

This project provides practical experience with AWS VPC design, subnet segmentation, routing, internet and NAT gateways, bastion-based administration, private database access, and VPC peering. Keeping application and database resources in private subnets reduces direct exposure, while deliberately configured routes and access rules support necessary communication. The submitted screenshots preserve evidence of the setup and validation workflows.

Operational consideration: NAT Gateways and other AWS resources may incur charges. Review and remove resources that are no longer needed, and protect SSH keys and credentials.